Sean Wang, Ph.D. (Chun-Hsiang Wang)

Plant Pathologist & Applied Research Scientist | Crop Protection, Seed Treatment & Pathogen Diagnostics
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PROFESSIONAL SUMMARY

Plant pathologist and applied research scientist (Ph.D., 2026), now a Post-Doctoral Associate at the UF/IFAS North Florida Research and Education Center (NFREC), with 7+ years of laboratory, greenhouse, and field research in crop protection, seed treatment, biologicals, pathogen diagnostics, and applied disease management. Experienced in pathogen isolation, inoculation, disease evaluation, EC_{50} sensitivity screening, seed treatment evaluation, field efficacy trials, and AI-assisted research workflows (Codex, Claude Code). Lead author of a peer-reviewed field efficacy study on fungicide spraying programs for carrot disease management, with additional experience preparing USDA-NIFA OREI and CPPM proposals and translating research data into practical crop-protection decisions.

CORE COMPETENCIES

- **Plant Pathology & Diagnostics:** Fungal and oomycete pathogen isolation, culture and inoculation · Disease ID (fungal, oomycete, bacterial, viral) · PCR / RT-PCR · Multi-locus phylogenetics and network analysis (R)
- **Field, Greenhouse & Efficacy Trials:** EC_{50} fungicide sensitivity profiling · Multi-year field and greenhouse trials · RCBD / CRD design · AUDPC, disease and crop-safety assessment · Treatment performance comparison · Trial operations, applications, data collection and technical reporting
- **Seed Treatment & Seed Health:** Germination, emergence and vigor evaluation · Seed-borne disease suppression · Seed-treatment performance · Seed safety / phytotoxicity assessment · Designed ISTA-aligned seed testing and germination protocols
- **Insect Management Research:** Applied IPM research on tea mosquito bug (*Helopeltis fasciaticollis*) · Host-preference and feeding-behavior observations · Insect handling · Injury-pattern documentation · Pest management data collection
- **Product Development & Crop Protection:** Antagonistic-microbe screening · *Bacillus* fermentation and biocontrol scale-up · Conventional, biological and botanical crop-protection products · Efficacy data supporting product positioning

PROFESSIONAL & RESEARCH EXPERIENCE

Post-Doctoral Associate | University of Florida (NFREC) *Quincy, FL · 2026 – Present*

Applied and translational research on a three-year USDA-NIFA OREI project on *Alternaria* leaf blight suppression in certified-organic carrot production (2026–2029; PI: Dr. Sanju Kunwar).

- Designated lead for day-to-day field trials and NOP-compliant seed phytosanitation.

Graduate Research Assistant | University of Florida *Gainesville, FL · 2022 – 2026*

Led applied plant pathology research on carrot diseases, especially *Alternaria* leaf blight, integrating field, greenhouse and laboratory work with seed-treatment evaluation, pathogen diagnostics, phylogenetic analysis, disease assessment and data analysis to support practical disease-management decisions.

- Designed and coordinated multi-year field experiments across conventional, organic and regenerative production systems, including ~3-acre fungicide efficacy trials whose field-trial results helped California carrot producers, including Grimmway Farms, design practical spray programs.
- Performed disease assessment, crop-health evaluation, AUDPC analysis, crop-safety observations and treatment-performance comparisons across application methods and products.
- Worked with fungal and oomycete pathogens, inoculation protocols and disease-rating systems for field-based efficacy evaluation; diagnosed *Pythium* contamination in packing facilities to support corrective action.

- Conducted pathogen identification and multi-locus phylogenetic analyses, surveying 84 isolates and establishing *A. alternata* as the dominant carrot ALB pathogen in Florida, to support diagnosis, isolate characterization and applied management research.
- Evaluated seed treatments and seed health, including germination, emergence, vigor, disease suppression, treatment performance and crop-safety assessment under controlled and applied research conditions.
- Collaborated with the Wisconsin resistance breeding team and California & Florida carrot producers to connect research findings with practical crop-production and crop-protection needs.
- Managed research timelines, data collection, technical documentation, scientific reporting and presentations for academic, grower and industry audiences.

Technical Extension Specialist & Researcher | Ministry of Agriculture *Hualien, Taiwan · 2021 – 2022*

- Applied IPM programs: Developed and implemented integrated disease-management protocols for Welsh onion and watermelon that reduced grower pesticide use by 80–90% while maintaining marketable yield.
- Biocontrol and diagnostics: Optimized *Bacillus* fermentation for biological control, screened antagonistic microbes, advanced lab candidates toward field application and tracked viral outbreaks by RT-PCR.

Graduate Research Assistant | National Taiwan University *Taipei, Taiwan · 2018 – 2020*

- *In vitro* screening and diagnostics: Conducted fungicide sensitivity assays and phylogenetic characterization of *Stemphylium* leaf blight on Welsh onion, building the molecular-diagnostics foundation later applied to field management programs.

IPM Research Intern | Ministry of Agriculture, Taiwan *Nantou, Taiwan · Jul 2016*

- Applied IPM research: Supported applied IPM research on tea mosquito bug (*Helopeltis fasciaticollis*) in tea production systems; conducted host-preference and feeding-behavior observations, insect handling, injury-pattern documentation, and pest management data collection.

GRANTS & FUNDING

- **Genotype × Management (G×M) Strategies and Epidemiological Forecasting for *Alternaria* Leaf Blight Suppression in Certified-Organic Carrot Production.** USDA-NIFA Organic Agriculture Research and Extension Initiative (OREI), Tier 3 Integrated. Key Personnel (Post-Doctoral Associate)—wrote the project narrative and generated the preliminary data; designated to manage day-to-day field trials and NOP-compliant seed phytosanitation. Funded 2026.

SELECTED PRESENTATIONS

- Wang, C.-H. (presenter), Da Silva, S., Hochmuth, R., Dufault, N., Rollins, J., Nakazawa, P., Maltais Landry, G., Treadwell, D., Dittmar, P., & Paret, M. (2026). *Alternaria* Leaf Blight Management in Florida Carrots: Pathogen Characterization, Fungicide Efficacy, and Regenerative Farming Strategies. *Oral presentation, 42nd International Carrot Conference, Bakersfield, CA, June 2026*.
- Wang, C.-H. (presenter), Nakazawa, P., Da Silva, S., Maltais Landry, G., Treadwell, D., & Paret, M. (2026). Assessing the Impact of Regenerative Agriculture Practices on Foliar Disease Severity in an Organic Crop Rotation System in Florida. *Oral presentation, Southern Division–American Phytopathological Society (SD-APS) Annual Meeting, Tifton, GA, March 2026*.

EDUCATION

- **Ph.D. in Plant Pathology**, University of Florida — 2026
- **M.S., Plant Pathology**, National Taiwan University — 2020
- **B.S., Plant Pathology**, National Chung Hsing University — 2018
- **High School,** National Taichung First Senior High School— 2014

CROP & PATHOGEN EXPERIENCE

- **Crops:** Carrot, peanut, grain corn, cover crops, Welsh onion, watermelon, cabbage, chayote, tea.

- **Pathogens & pests:** *Alternaria* leaf blight, *Sclerotinia* cottony rot, *Pythium* cavity spot, *Stemphylium* leaf blight, peanut leaf spot, corn ear rot, watermelon gummy stem blight, anthracnose, viral diseases, tea mosquito bug.

SELECTED PUBLICATIONS

- Wang, C.-H., Silva, S.D., Hochmuth, R., Hurley, D., Broughton, B., Dufault, N., et al. (2026). Efficacy of fungicides applied via solid-set overhead sprinklers vs. tractor-mounted sprayers for carrot *Alternaria* leaf blight. *Crop Protection*.
- Wang, C.-H. et al. (2026). Management of *Alternaria* Leaf Blight of Carrot for Florida Conventional and Organic Producers. *UF/IFAS EDIS, PP392*.
- Wang, C.-H., Tsai, Y.-C., Tsai, I., Chung, C.-L., Lin, Y.-C., Hung, T.-H., Suwannarach, N., et al. (2021). *Stemphylium* leaf blight of Welsh onion (*Allium fistulosum*): an emerging disease in Sanxing, Taiwan. *Plant Disease, 105(12), 4121–4131*.
- Ketehouli, T., Okpala, E.N., Wang, C.-H., et al. (2026). Harnessing plant ferritins for iron homeostasis and crop resilience: an overview. *Plant Stress*.
- Wu, C.-Y., Tsai, Y.-C., & Wang, C.-H. (2023). Preliminary study on screening antagonistic microorganisms for controlling pepper anthracnose and cucurbit gummy stem blight. *Annual Meeting, Plant Protection Society of the R.O.C.*